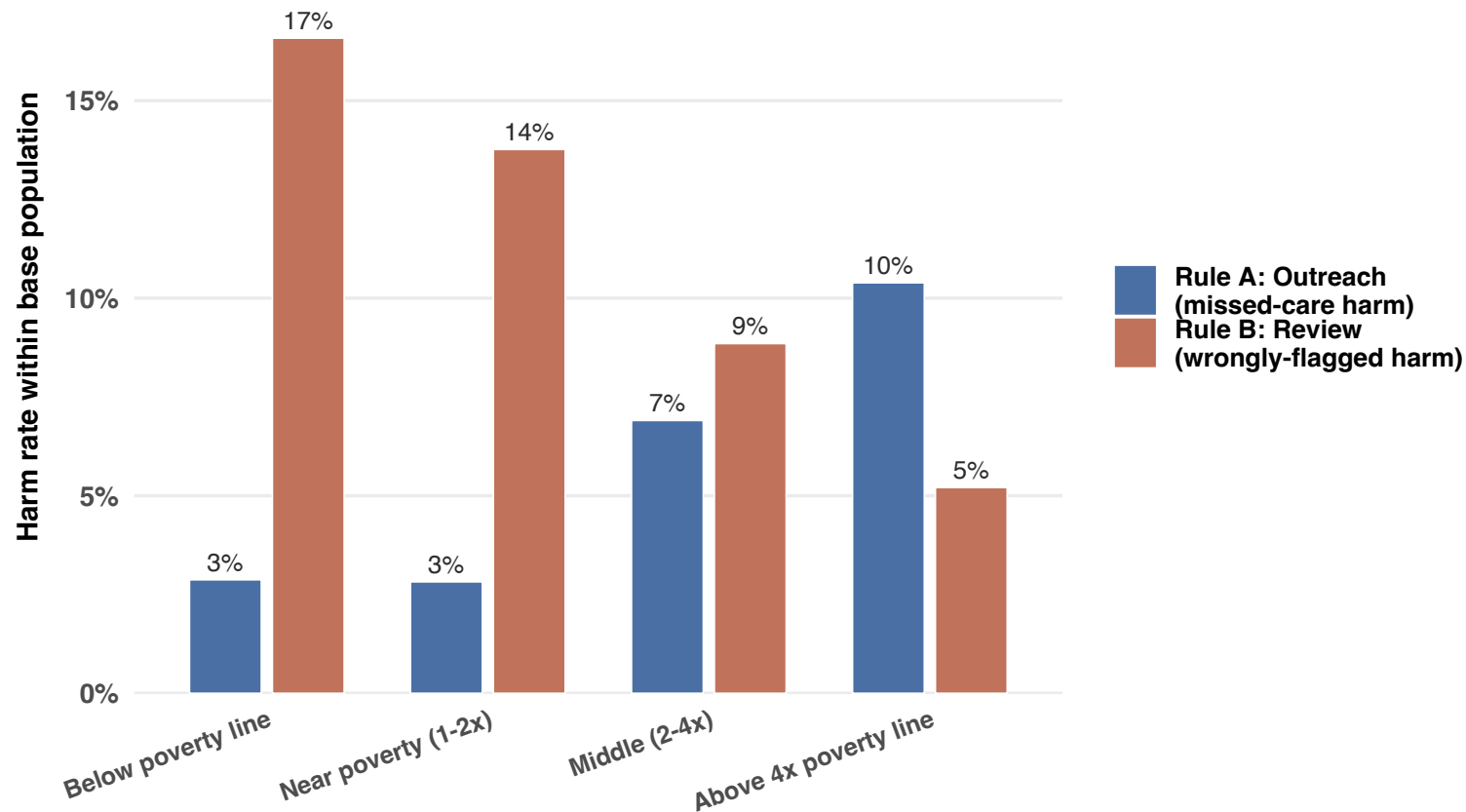


Who the proxy harms depends on what it's used for

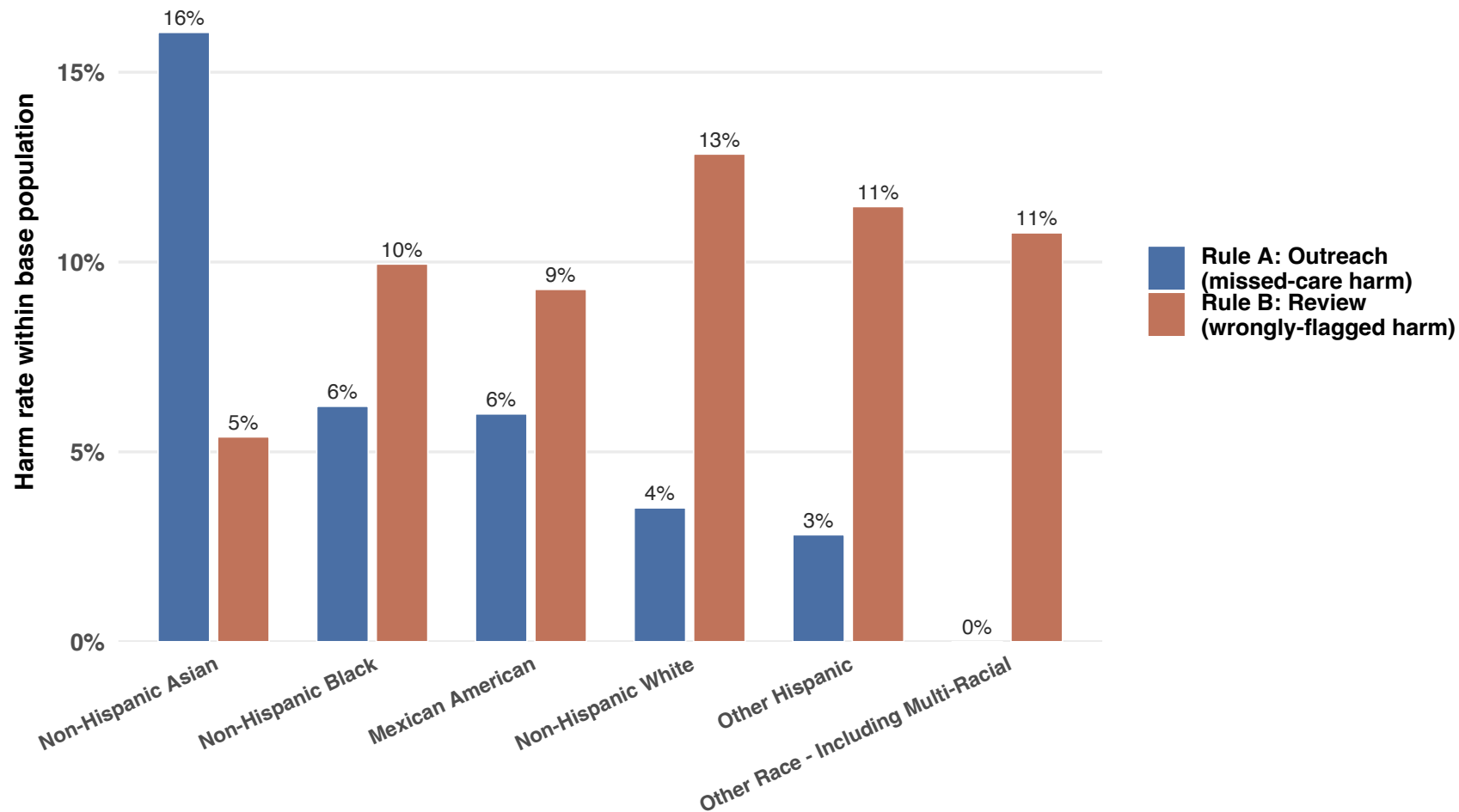
Wealthier disabled respondents are missed by outreach; poorer non-disabled respondents are wrongly flagged for review -- same model, opposite populations harmed



Base population: true disability cases (Rule A) / true non-disability cases (Rule B).
NHANES 2013-2018 pooled, n = 4,733 held-out test respondents.

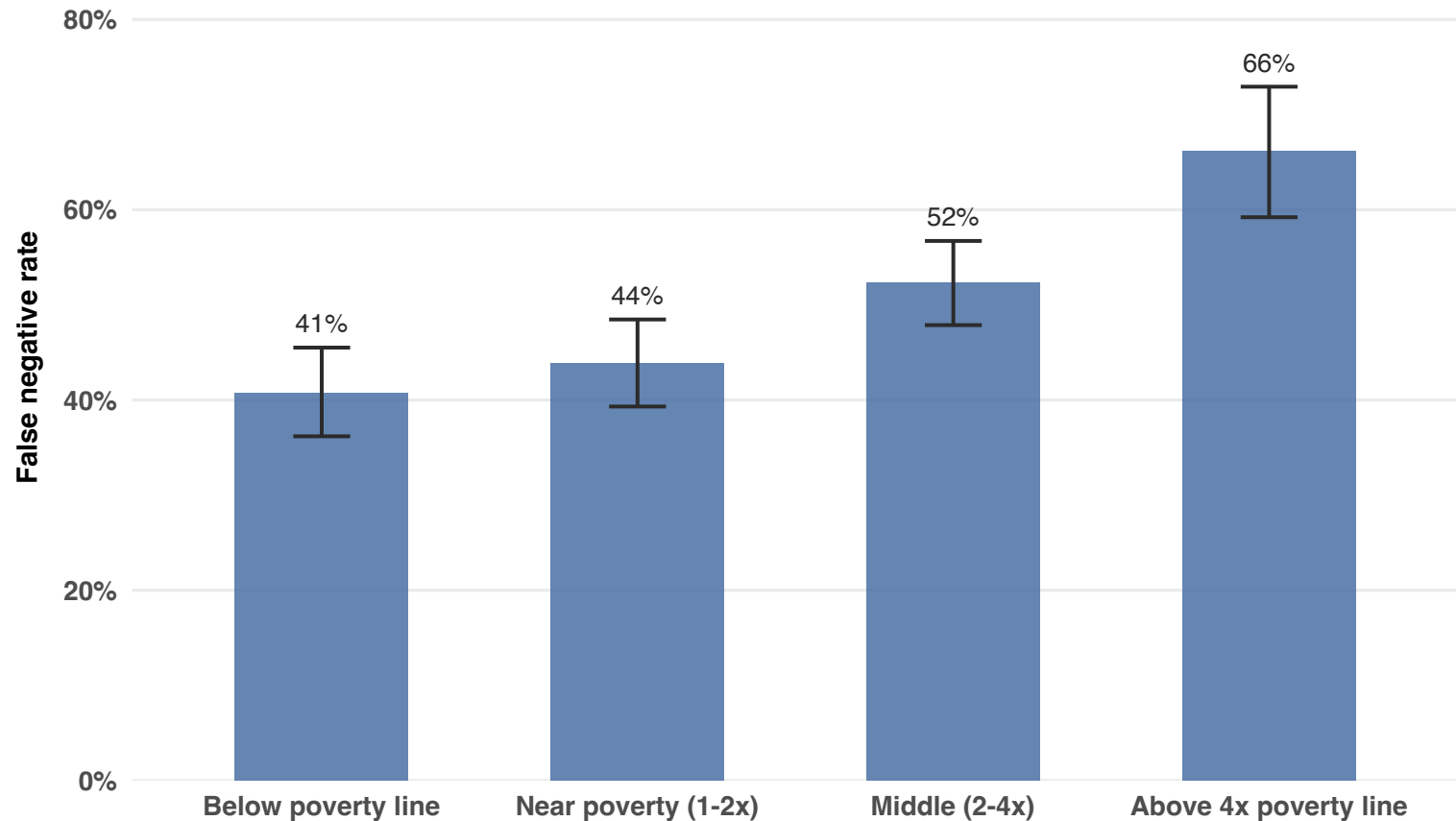
The same inversion holds across race/ethnicity

Non-Hispanic Asian respondents are most missed by outreach but least likely to be wrongly flagged for review -- White respondents show the reverse



The income gap is statistically real, not sampling noise

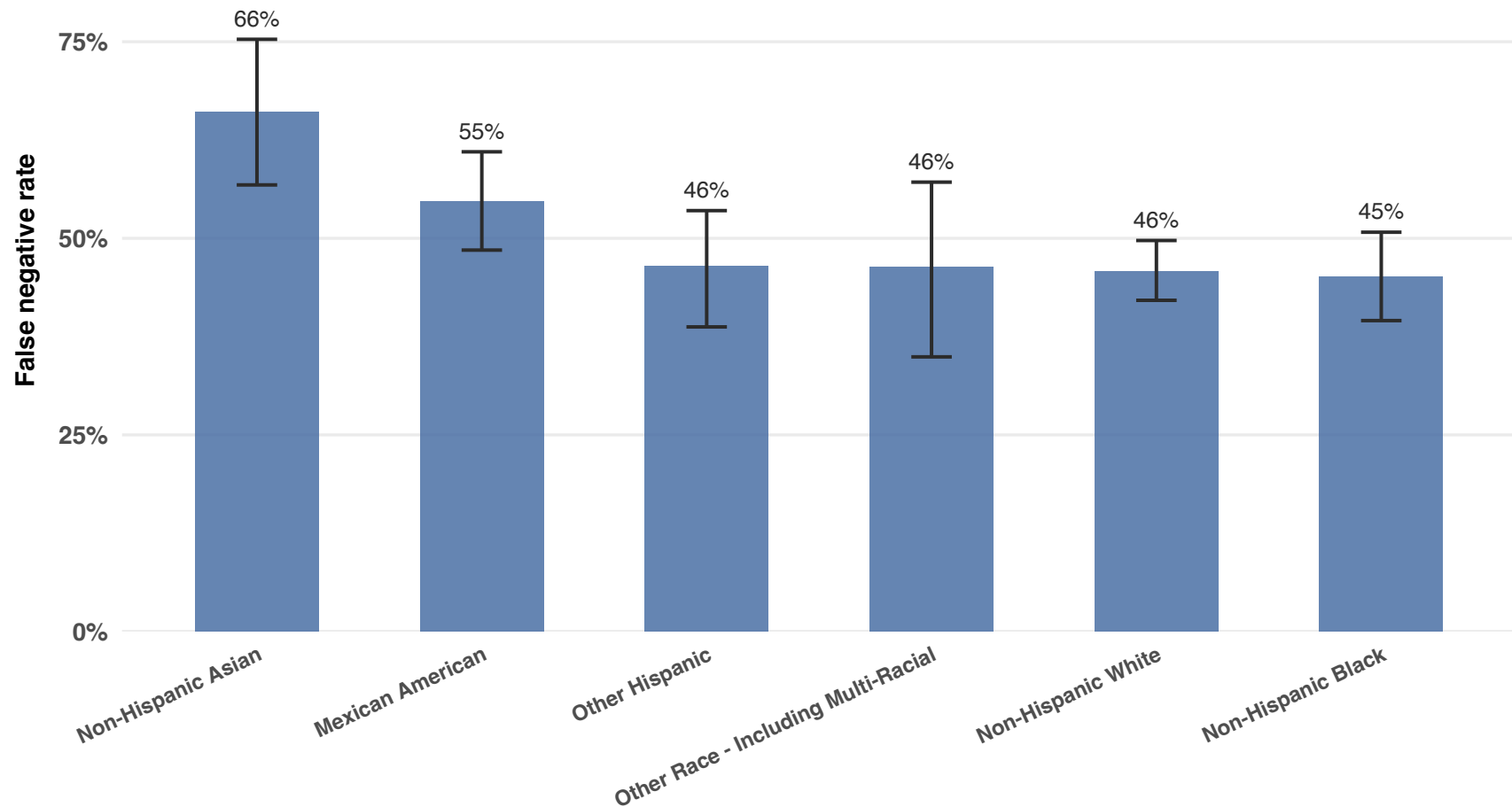
False-negative rate by income tier, with 95% bootstrap confidence intervals (2,000 iterations) -- the poorest and wealthiest tiers' intervals do not overlap



Bootstrapped 95% CIs, 2,000 resampled model refits on pooled NHANES 2013-2018 data.

Non-Hispanic Asian respondents: the clearest race-based gap

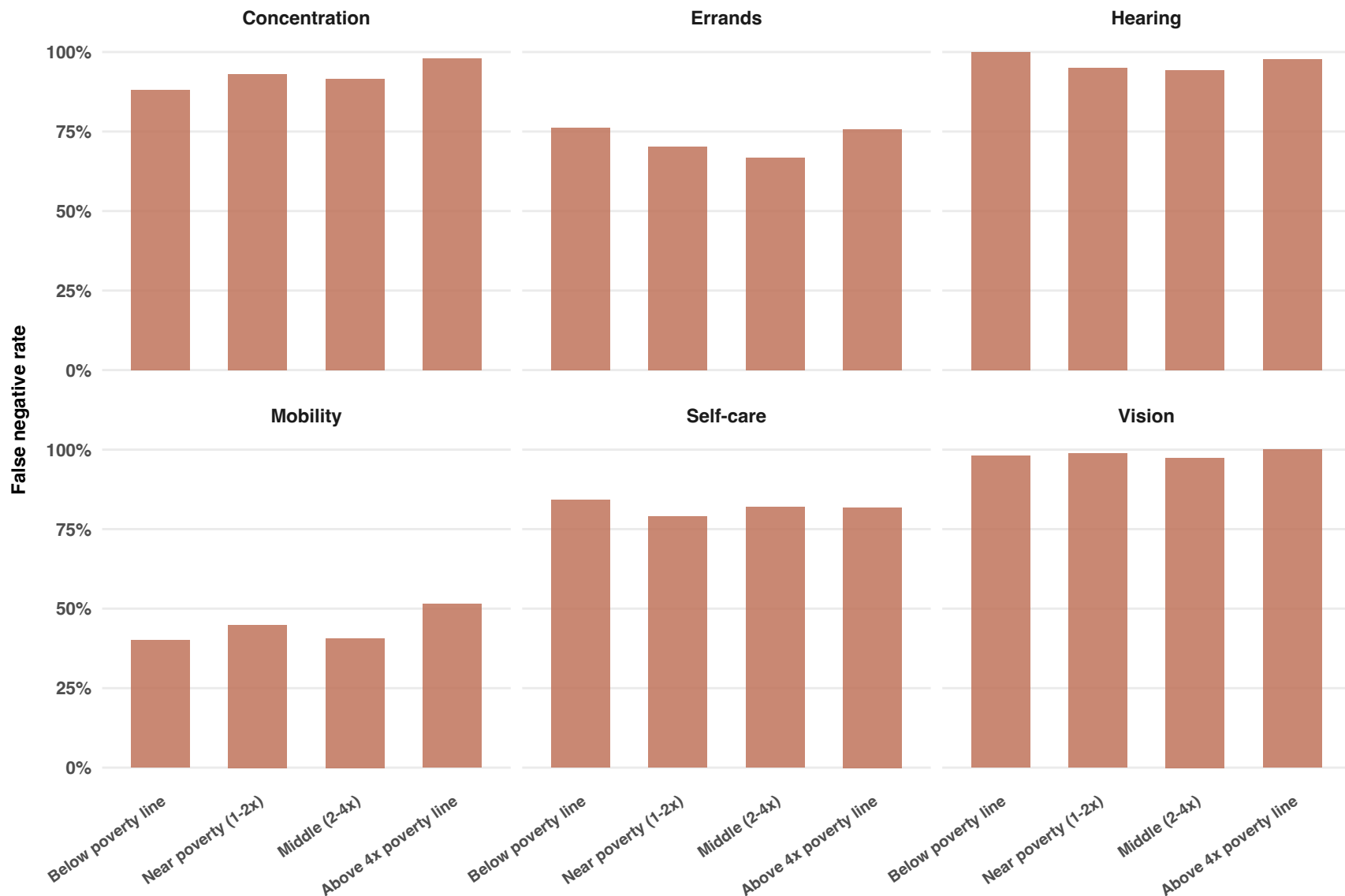
False-negative rate by race/ethnicity, with 95% bootstrap confidence intervals --
Asian and White respondents' intervals no longer overlap after pooling cycles



Bootstrapped 95% CIs, 2,000 resampled model refits on pooled NHANES 2013-2018 data.

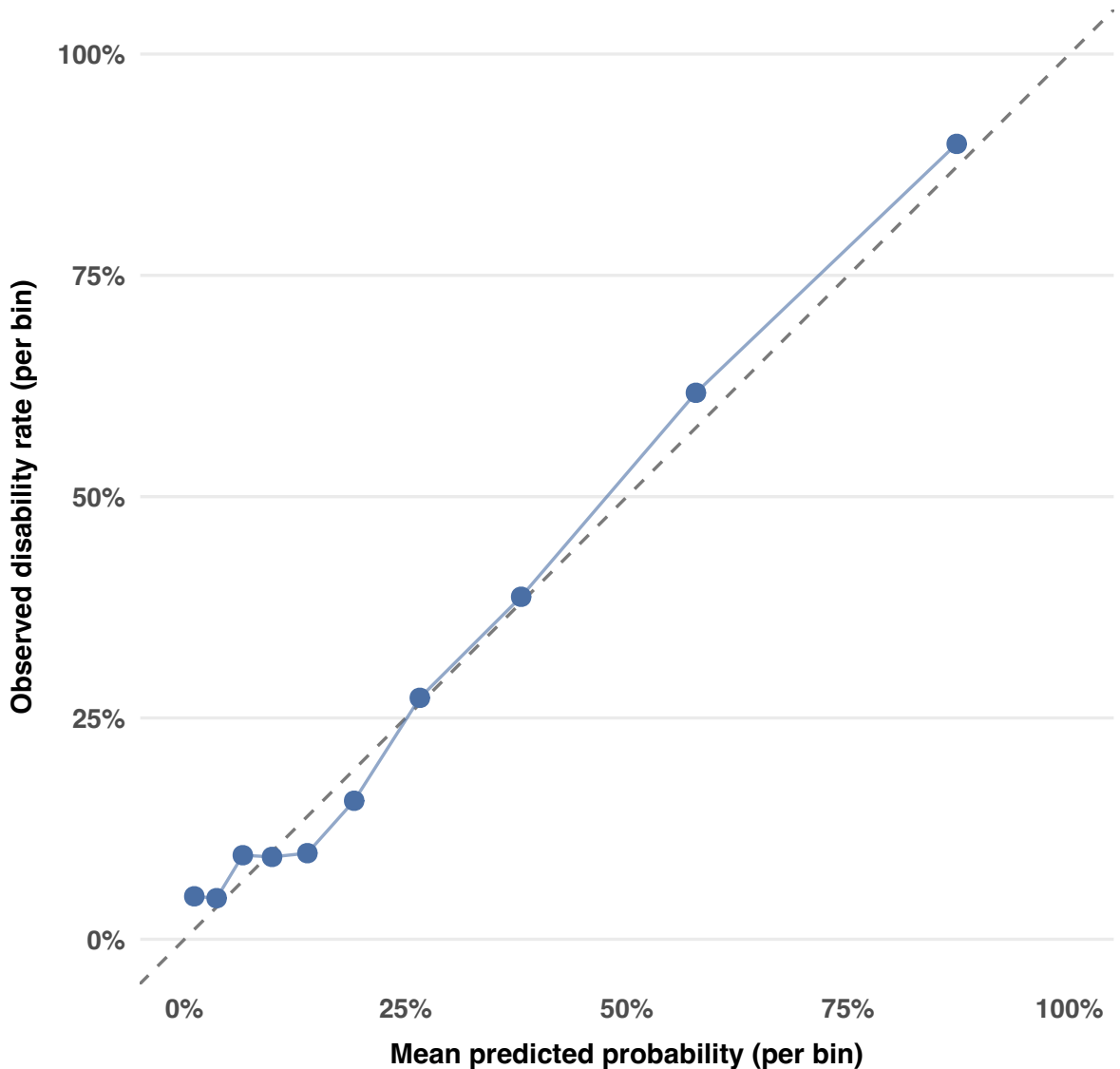
The income gradient is concentrated in mobility, not other domains

Hearing and vision difficulty are barely detectable by this proxy for ANY income group; mobility is the one domain with real signal -- and the clearest income gradient



Predicted probabilities track observed rates well overall

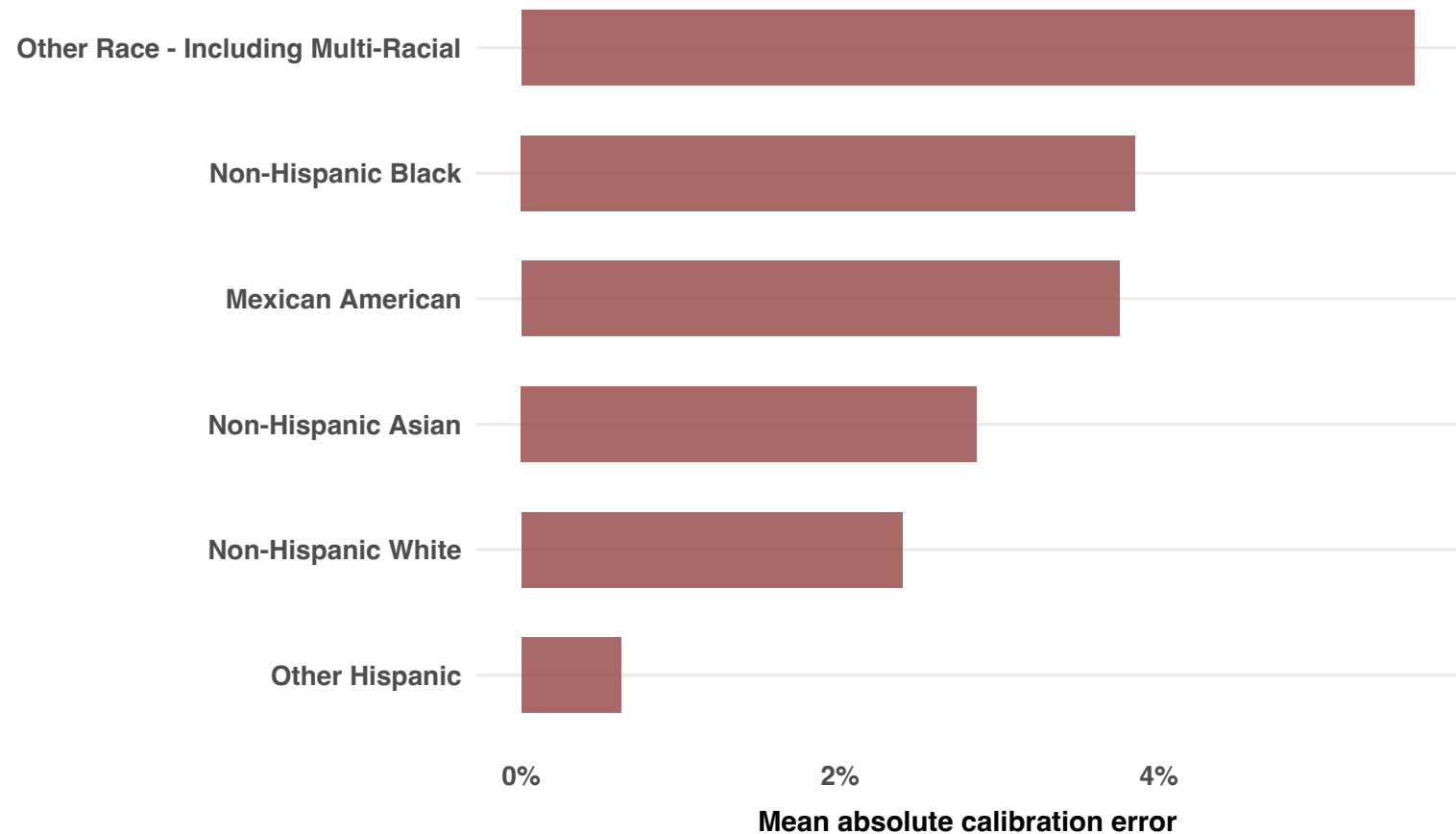
10-bin calibration curve, all respondents pooled -- points close to the dashed diagonal indicate predicted probabilities are reasonably trustworthy on average



Mean absolute calibration gap (overall): 0.024. Pooled NHANES 2013-2018 held-out test set.

Calibration error and sensitivity gaps don't track each other

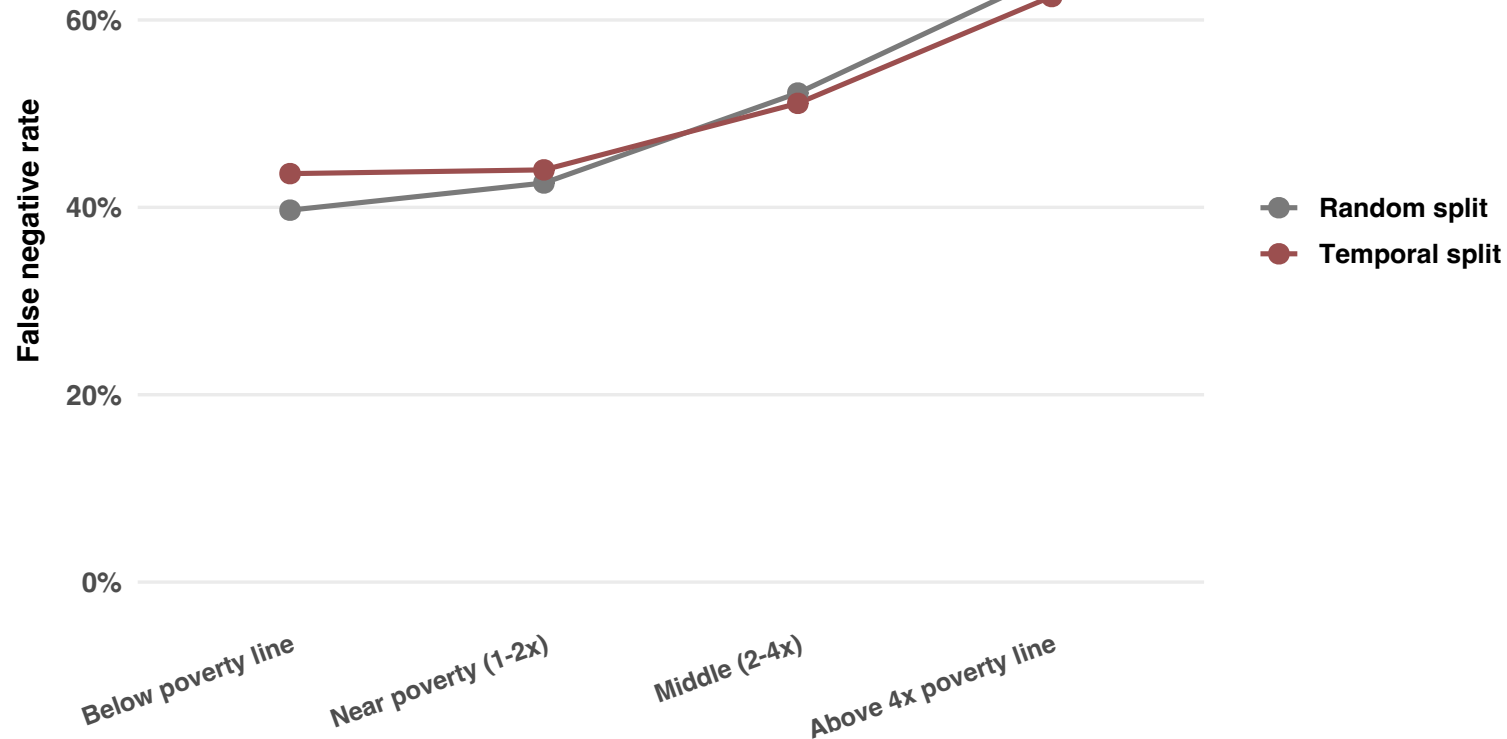
Non-Hispanic Asian respondents have the worst SENSITIVITY but reasonably good CALIBRATION; "Other Race/Multiracial" shows the opposite pattern



Lower = predicted probabilities track observed outcomes more closely for that group.
Pooled NHANES 2013-2018.

The income gradient survives the strictest possible test

Training on 2013-2016 and testing ONLY on fully unseen 2017-2018 respondents produces nearly identical results to a random pooled split



Random split: Step 3d (train+test mixed across cycles). Temporal split: Step 4d (train on cycles H+I only, test exclusively on cycle J).